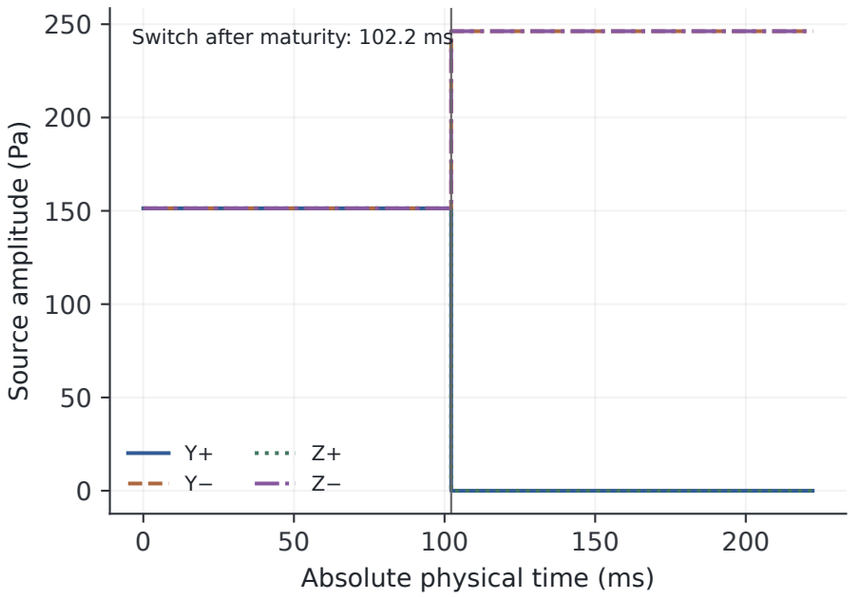
F2-3 | PZT control of an already mature cluster
Same 102.2 ms state · 120 ms continuation · actual basis field · EXPLORATORY

a Amplitude / enable control



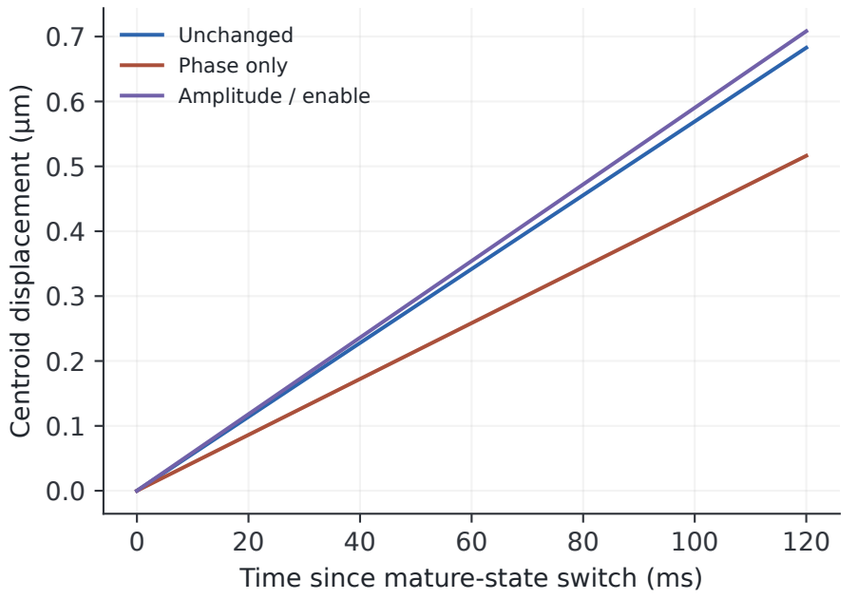
b Phase-only control

Y+ / Y− / Z+ / Z−
0 / 0 / 0 / 0 → 30.0 / 45.67 / 0 / 0 deg

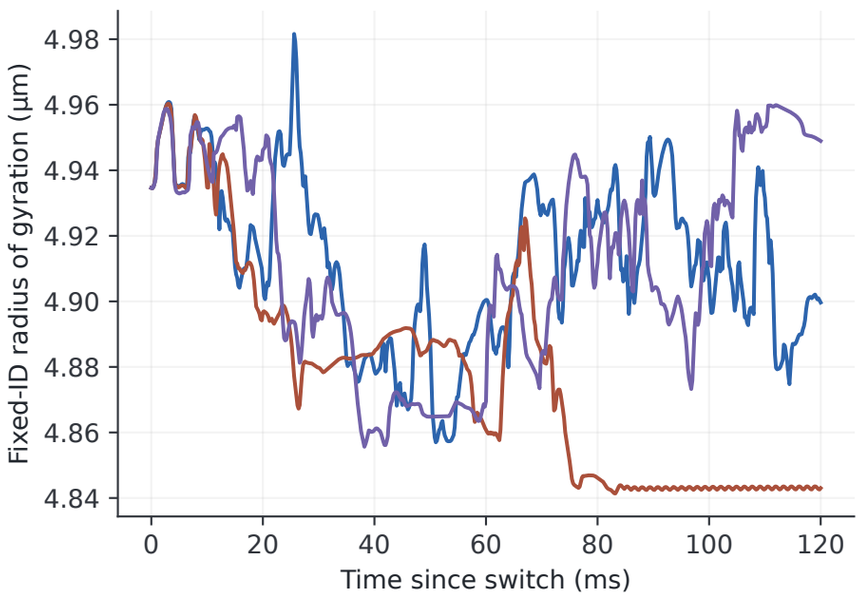
Amplitudes unchanged: 151.34 Pa each
All four elements remain enabled
ROI pressure RMS matched: 843.77 Pa
27 fixed diagnostic points; all controls

Same mature state: 32 / 32 bubbles
10 ms stable membership before switch

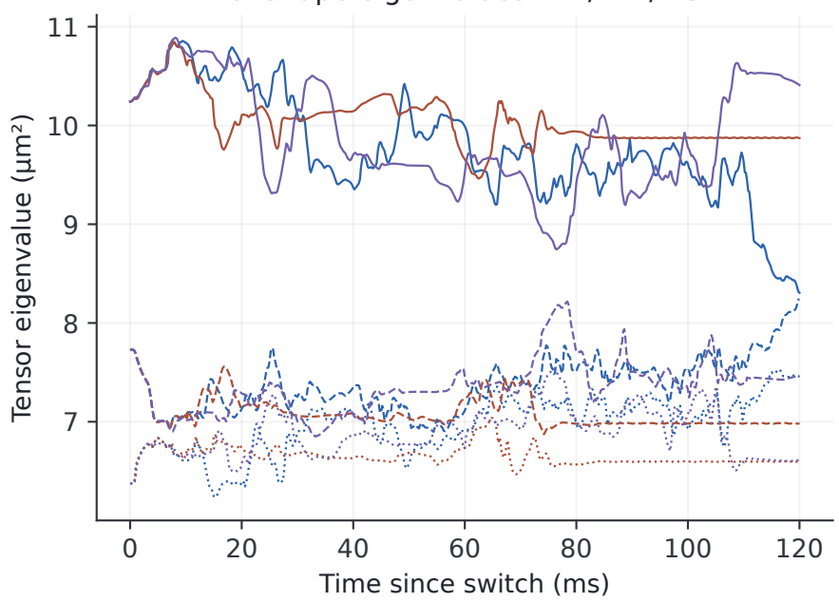
c Translation from mature state



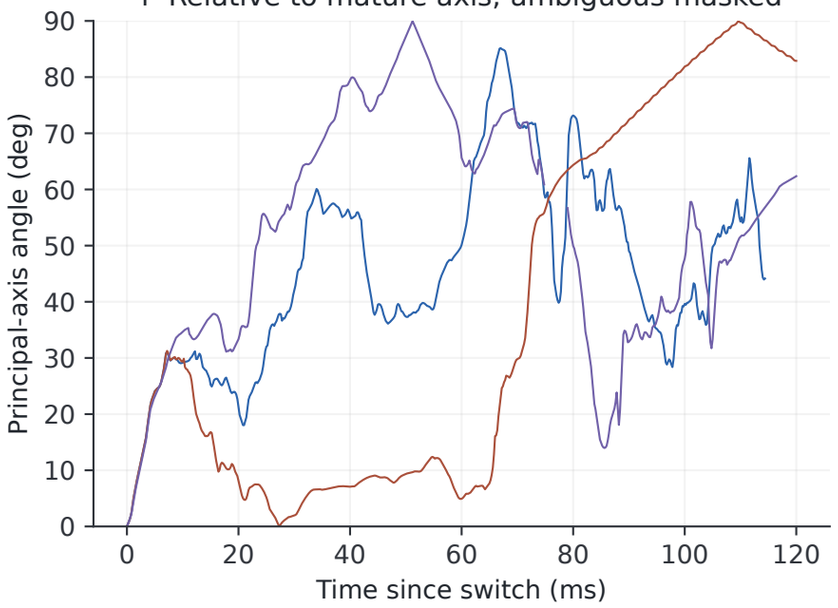
d Overall size



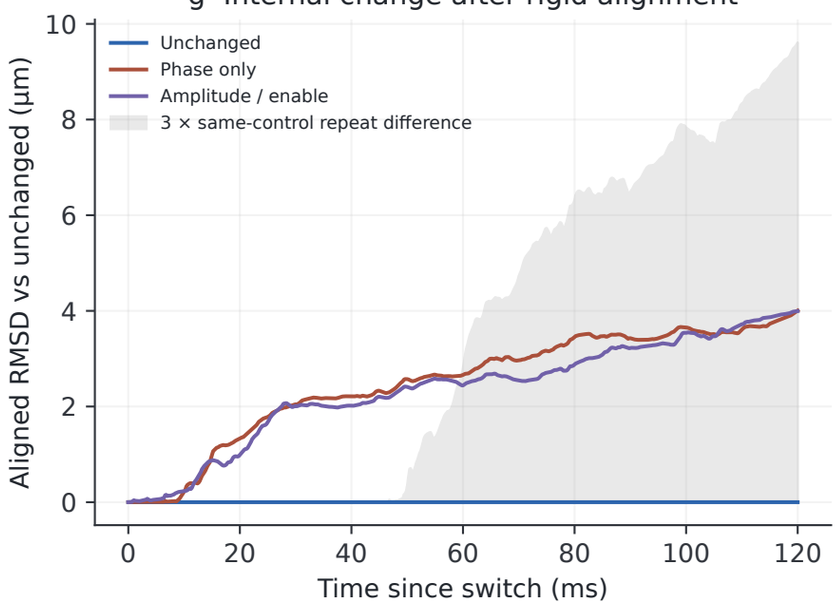
e Shape eigenvalues: λ1 / λ2 / λ3



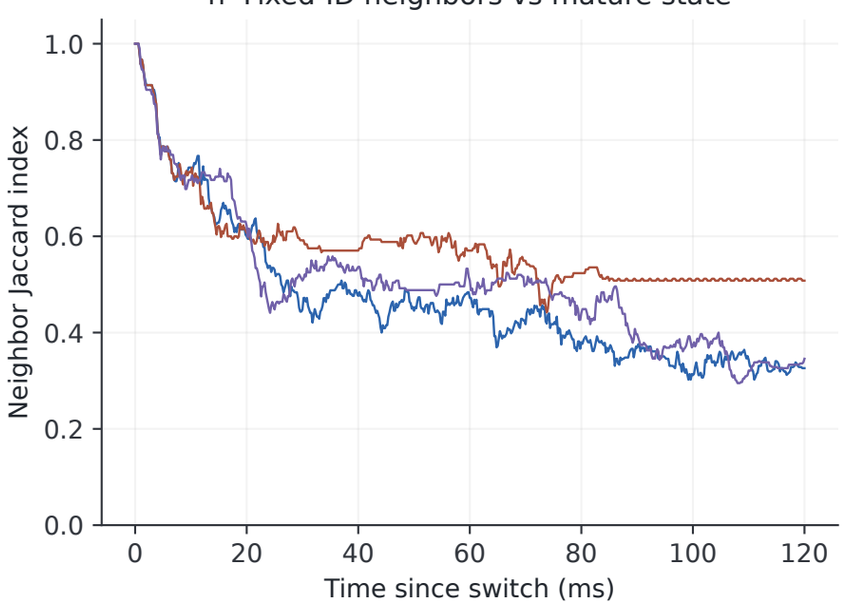
f Relative to mature axis; ambiguous masked



g Internal change after rigid alignment



h Fixed-ID neighbors vs mature state



i Topology and inference

All branches: one 32-bubble component
Persistent merge / split: 0 / 0
Transient component merge / split: 0 / 0
Membership exchange: none

Raw pair-edge changes:
unchanged 1479; phase 633; amplitude 979
These do not change component membership.

NO_SIGNIFICANT_SHAPE_RESPONSE
Endpoint shape effects do not exceed
the registered repeat envelope.
Transient eigenvalue windows: up to
5.4 ms (amplitude), 5.6 ms (phase).
One repeat; descriptive evidence only.